

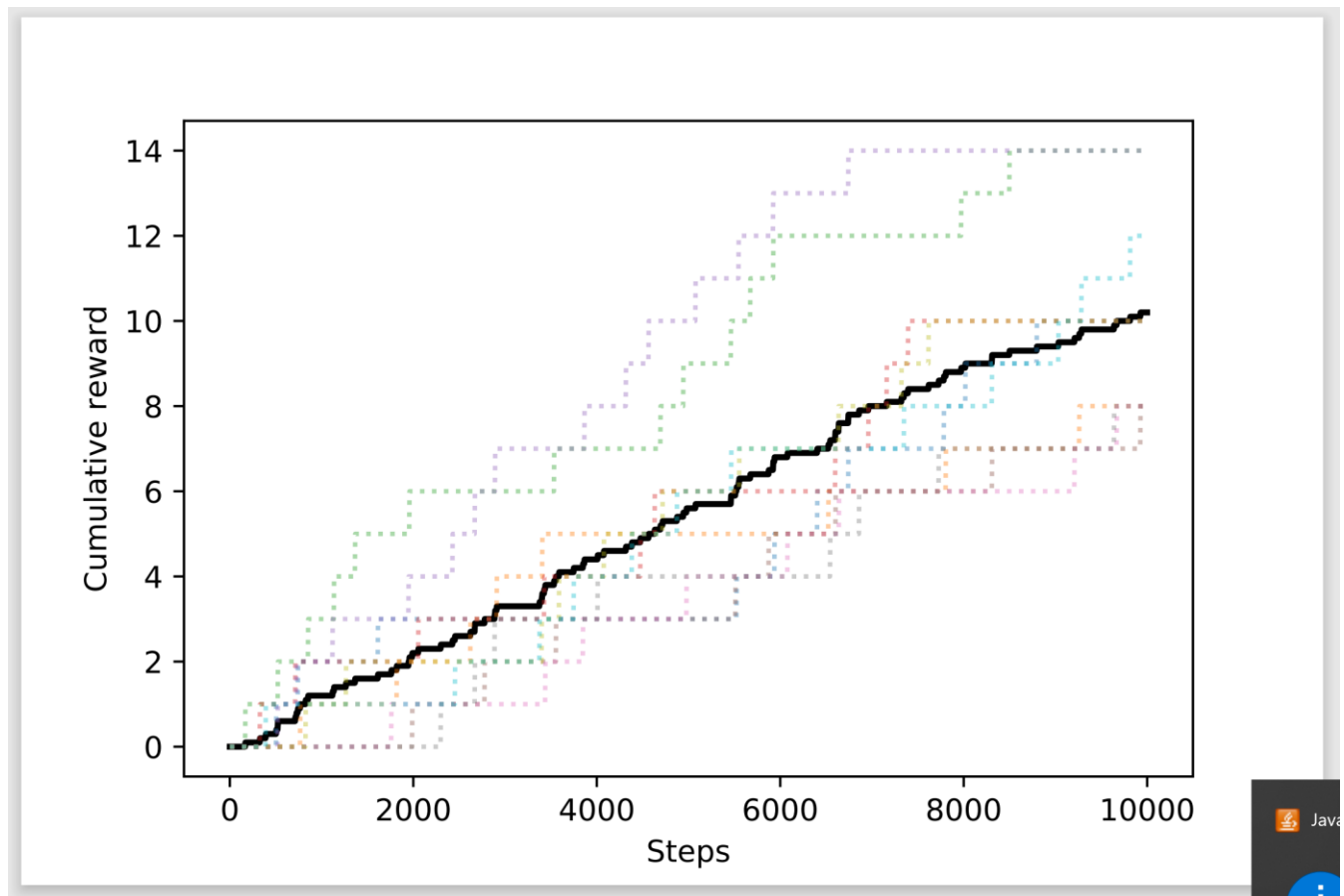
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Exercise 0: An Invitation to Reinforcement Learning

3. The Manual performance should be an improvement on the random policy. The manual policy asks the user to enter an action, and the user will choose the most direct action to take (plus the 10% chance of perpendicular motion.) The random policy can eventually make it to the goal, but it takes unnecessary actions that will not take it towards the goal. So, the random policy should on average have more actions per reward than the manual policy.



4. The random policy selects one of the four actions with equal likelihood. The worse policy, it is very similar to the random policy, but it only goes in the up or right direction with equal likelihood. The better policy, a sort of preferred random strategy, it will change what actions are available and the distribution of probabilities for each action available based on what quadrant of the graph it is in. The rules are:

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#if below horizontal wall, go up and try to not go left or right very often,  
#if above horizontal wall, go to the right and up slightly more than down,  
#if in upper right quadrant go up and to the right only with equal probability
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This set of instructions should take more direct actions than a truly random policy, but perhaps take more non-goal-oriented actions. Resulting in more actions per reward than a manual policy, and less actions per reward than a truly random policy.

